



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 73



KOHLER DIESEL ENGINE
Model : KD440

KOHLER KD440 Diesel Engine

Test Report No.	:	2013 – 73
Date of Test	:	June 4 - 5, 2013
Test Valid Until	:	June 5, 2018
Brand	:	KOHLER
Model	:	KD440
Serial No.	:	5946878
Type	:	Four-stroke, air-cooled, vertical, single-cylinder diesel engine
Manufacturer	:	Made in Italy
Test Requested by	:	DKI CORPORATION 112 Amorsolo St., Legaspi Village, Makati City, Metro Manila
Machine Classification	:	Commercial

SUMMARY

The KOHLER KD440 Diesel Engine was tested to establish its performance. It was tested at the full throttle speed setting pre-set by the manufacturer / test applicant.

The Maximum Power developed by the engine was 6.34 kW at 3706 rpm Engine Shaft Speed. At Maximum Power, the Torque on the output shaft of the engine was 1.67 kg-m with Fuel Consumption Rate of 2.27 L/h. The Noise Level measured at this condition was 91.5 db (A). The Maximum Torque developed was 1.87 kg-m recorded from the Engine Shaft Speed of 3107 rpm.

For the Continuous Running Test, the engine was set at the Maximum Power it can attain in its Rated Shaft Speed. The Average Output Power for the five-hour test was 6.10 kW at 3616 rpm Engine Shaft Speed. This was 79.3% of the Rated Maximum Power.

The engine was easy to start. It only took a single attempt to start the engine using rope-recoil in both cold and hot starting.

SCOPE OF TEST

The engine was tested to establish its performance characteristics. The test was conducted from June 4 to 5, 2013 at the AMTEC Test Laboratory. It was evaluated on the following items:

- a. Suitability of operator's manual;
- b. Starting test;
- c. Varying load performance test; and
- d. Continuous running test

METHOD OF TEST

The engine was tested based on the Philippine Agricultural Engineering Standards, PAES 117:2000 Agricultural Machinery – Small Engines – Method of Test.

DESCRIPTION OF THE ENGINE

The KOHLER KD440 Diesel Engine (**Figure 1**) had a Rated Maximum Output Power of 7.69 kW (10.3 hp) at Rated Engine Shaft Speed of 3600 rpm, based on plate rating. It was a four-stroke, air-cooled, vertical, single-cylinder diesel engine. Fuel feed system was gravity type with the fuel tank located on top of the engine. It utilized oil bath type air cleaner. Lubrication system was splash type. Cooling was done by air forced by a fan/flywheel encased in a metal shroud. It was started using rope-recoil starting system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the engine.



Figure 1. KOHLER KD440 Diesel Engine.

ENGINE SPECIFICATIONS

Table 1. Detailed engine specifications of the KOHLER KD440 Diesel Engine.

1.	Dimensions and Weight		
1.1	Overall length (mm)	:	480
1.2	Overall width (mm)	:	395
1.3	Overall height (mm)	:	500
1.4	Weight (kg)	:	52
2.	Engine Specifications (based on plate rating, manufacturer's claims, and ocular inspection of the engine)		
2.1	Brand	:	KOHLER
2.2	Model	:	KD440
2.3	Serial No.	:	5946878
2.4	Make	:	Italy
2.5	Type	:	Four-stroke, air-cooled, single-cylinder, vertical diesel engine
2.6	Rated Output, kW (based on plate rating)		
	Maximum Power	:	7.69 kW @ 3600 rpm
	Continuous Power	:	No data available
2.7	Bore x stroke (mm)	:	86 x 76
2.8	Total displacement volume (cc)	:	442
2.9	Fuel system		
	2.9.1 Type of fuel feed	:	Gravity
	2.9.2 Fuel used	:	Diesel
2.10	Cooling system	:	Air-cooled
2.11	Lubricating system	:	Splash type
2.12	Starting system	:	Rope-recoil
2.13	Air cleaner	:	Oil bath type
2.14	Exhaust system	:	Muffler

OPERATOR'S MANUAL

The operator's manual submitted with the engine was presented in six languages including English. It contained the following: Safety precautions, Engine Specifications, Engine Identification, Features and Controls, Maintenance and Warranty.

RESULTS AND DISCUSSION

The KOHLER KD440 Diesel Engine was tested to establish its performance. The test was conducted by directly coupling the power output shaft of the engine to the electric dynamometer. **Figure 2** shows the test set-up.



Figure 2. Test set-up of the KOHLER KD440 Diesel Engine.

A. STARTING TEST

Table 2 shows the result of the Starting test. The engine was started using the recommended starting procedure by the manufacturer. For cold starting, the engine was not operated for at least 12 hours and then started, while for hot starting, the engine was operated for one (1) hour and then stopped and re-started. The number of attempts needed to start the engine was recorded in both tests.

During cold and hot starting, it only took a single attempt to start the engine. In general, the engine was easy to start.

STARTING TEST

Engine on Test : KOHLER
Model : KD440
Date of Test : June 4, 2013

Table 2. Results of starting test.

ITEMS	COLD STARTING	HOT STARTING
1. Ambient temperature (°C)	32.0	
2. Relative Humidity (%)	74.7	
3. Engine oil temperature (°C)	30.0	77.0
4. No. of attempts to start the engine	1	1
5. Ease of starting	EASY	EASY

B. VARYING LOAD PERFORMANCE TEST

The engine mounted on the electric dynamometer was tested for varying load performance. Data were obtained at the recommended maximum no-load speed setting (Pre-set) of the engine and at prevailing ambient conditions during the time of test using locally available diesel fuel. **Table 3** and **Figure 3** show the results of the varying load performance test.

OUTPUT POWER. Based on the results of test, the Maximum Power developed by the engine, was 6.34 kW at 3706 rpm engine shaft speed. At Maximum Power, the Torque on the output shaft of the engine was 1.67 kg-m with Fuel Consumption Rate of 2.27 L/h. The Maximum Torque developed was 1.87 kg-m at an Engine Shaft Speed of 3107 rpm. The Actual Maximum Power was 82.4% of the Rated Maximum Power of the engine.

FUEL CONSUMPTION. The Fuel Consumption of the engine was obtained using ONO SOKKI Digital Fuel Consumption Meter Model: FC-014/024 coupled to an ONO SOKKI Burette Stand Model: PP-523. The Maximum Fuel Consumption Rate of the engine was 2.27 L/h. It occurred at the point of Maximum Output Power. The Specific Fuel Consumption at this point was 260.2 g/kW-h.

TEMPERATURES. Temperatures were measured using Type J thermocouples attached to a YOKOGAWA Digital Thermometer, Type 2575 coupled to YOKOGAWA multi-channel switch, Type 2815. The Maximum Temperature of the engine oil was 123.5 °C which occurred at 5.72 kW Output Power of the engine and at 2998 rpm Engine Shaft Speed. For the exhaust air temperature at the outlet of the muffler, the Maximum Temperature was 435.5 °C which occurred at 6.28 kW Output of the engine at an Engine Shaft Speed of 3607 rpm.

NOISE LEVEL. Noise Level was the quantified degree of sound emitted by the engine during operation. It was measured using a CUSTOM Sound Level Meter Model: SL-1370. The Noise Level of the engine was measured at 7.5 meters away from the muffler, 1.2 meters high from the ground and 45° from the engine shaft. The Maximum Noise Level measured was 91.5 db (A).

OBSERVATIONS. No breakdown and other malfunctioning occurred during the test.

VARYING LOAD PERFORMANCE TEST

Engine on Test : KOHLER
Model : KD440
Date of Test : June 4, 2013

Test Conditions

Ambient Temperature (°C) : 32.0
Relative humidity (%) : 74.7

Table 3. Results of varying load performance test¹

SHAFT TORQUE (kg-m)	ENGINE SHAFT SPEED (rpm)	OUTPUT POWER (kW)	FUEL CONSUMP- TION (L/h)	SPECIFIC FUEL CONSUMP- TION (g/kW-h)	TEMPERATURE (°C)		NOISE LEVEL [DB(A)]
					ENGINE OIL	EXHAUST AIR	
0.26	3847	1.01	0.96	692.3	77.0	192.0	90.4
1.07	3792	4.16	1.76	307.1	89.5	319.0	91.0
1.67	3706	6.34	2.27	260.2	98.0	421.5	91.5
1.70	3607	6.28	2.17	250.2	111.0	435.5	90.2
1.75	3505	6.28	2.08	240.5	115.5	425.0	89.3
1.77	3403	6.18	2.01	235.5	118.5	410.0	88.7
1.80	3309	6.12	1.95	231.7	120.0	396.0	87.7
1.86	3201	6.10	1.93	230.0	122.5	392.0	87.3
1.87	3107	5.96	1.88	228.1	122.0	382.0	87.1
1.86	2998	5.72	1.78	225.8	123.5	369.5	86.2
1.84	2793	5.28	1.60	219.5	123.0	355.0	85.5
1.84	2604	4.91	1.49	219.7	123.0	343.5	84.7
1.82	2399	4.48	1.35	217.9	123.0	321.5	83.8
1.75	2196	3.95	1.21	221.9	122.5	304.5	82.1
1.60	2002	3.29	1.00	220.9	121.5	284.5	80.3
1.48	1808	2.75	0.87	228.8	120.0	265.0	79.6
1.37	1598	2.25	0.74	237.5	119.0	240.5	78.5
1.23	1419	1.79	0.61	248.1	112.5	198.0	76.4
0.80	1203	0.99	0.45	329.8	110.5	175.5	74.4
0.53	971	0.52	0.30	409.5	109.0	182.0	74.1

¹ Average of three test trials

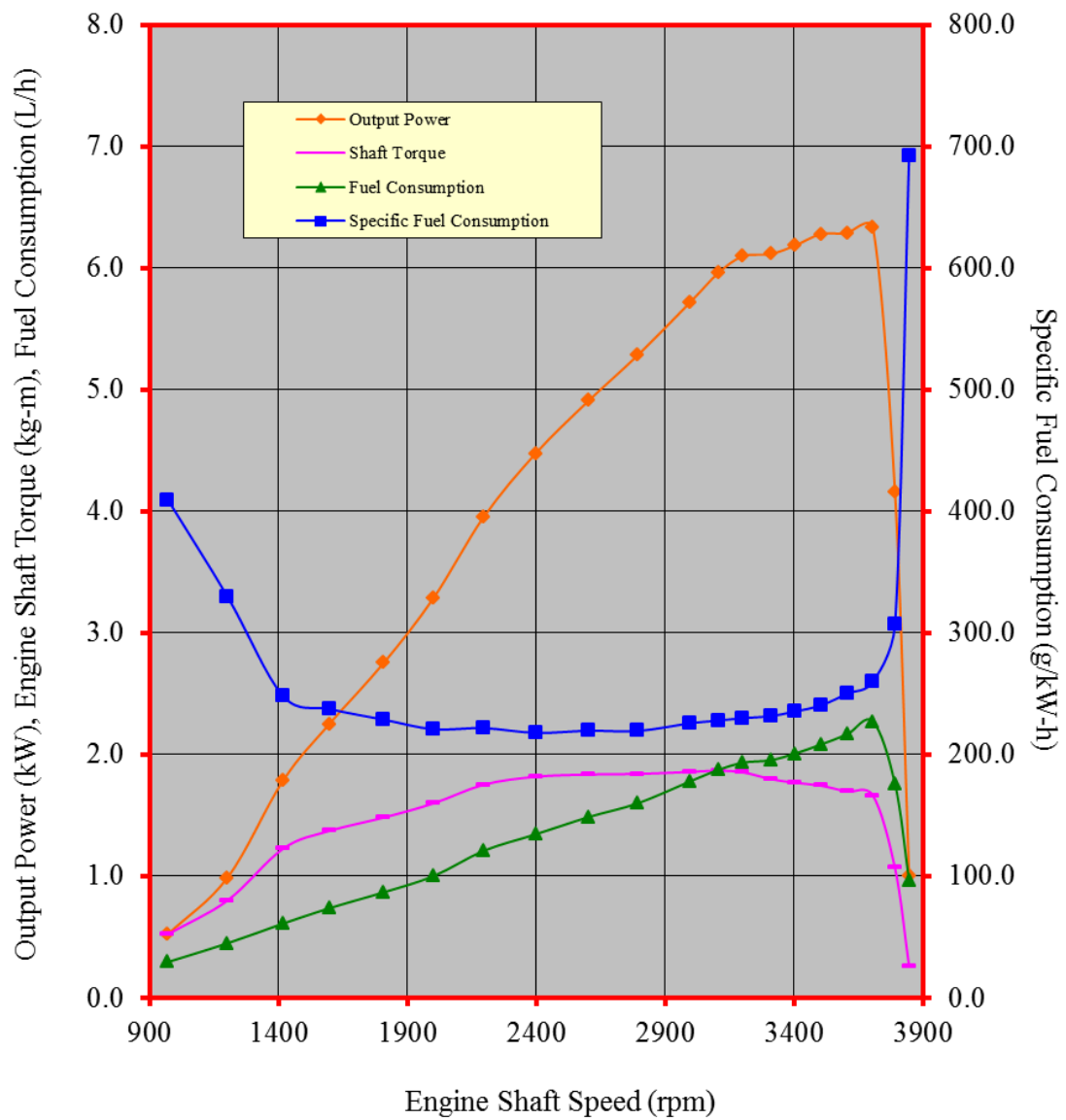


Figure 3. Performance characteristic curves of the KOHLER KD440 diesel engine

C. CONTINUOUS RUNNING TEST

The engine mounted on the electric dynamometer was made to run continuously for five hours. Initially, the throttle was set at the no-load speed setting as in the varying load test. As per standard procedure, the engine should be set at the Rated Continuous Power specified by the manufacturer. However, if not specified, the engine should be set at 80% of the Rated Maximum Power. If the engine cannot attain the 80% of the Rated Maximum Power specified by the manufacturer, it should be set at the Maximum Power it can attain at Rated Speed. As such, the engine was set at the Maximum Power it can attain in its Rated Shaft Speed. Data were gathered every thirty minutes. No breakdown occurred.

The Average Output Power obtained after the five-hour period was 6.1 kW at 3616 rpm Engine Shaft Speed. This was 79.3% of the Rated Maximum Output Power. Results of test are shown in **Table 4** and plotted in **Figure 4**.

CONTINUOUS RUNNING TEST

Engine on Test : KOHLER
 Model : KD440
 Date of Test : June 5, 2013
 Duration of Test (h) : 5

Test Conditions

Ambient Temperature (°C) : 32.6
 Relative humidity (%) : 73.2

Table 4. Results of continuous running test at 79.3% of the Rated Maximum Power.

Items Measured	Average values
Engine Speed (rpm)	3616
Torque (kg-m)	1.64
Output Power (kW)	6.10
Fuel Consumption (L/h)	2.13
Specific Fuel Consumption (g/kW-h)	253.6
Temperature (°C) :	
a) Engine Oil	136.6
b) Exhaust Gas	415.9
Noise Level [db(A)]	90.6

OBSERVATIONS/REMARKS: No breakdown occurred.

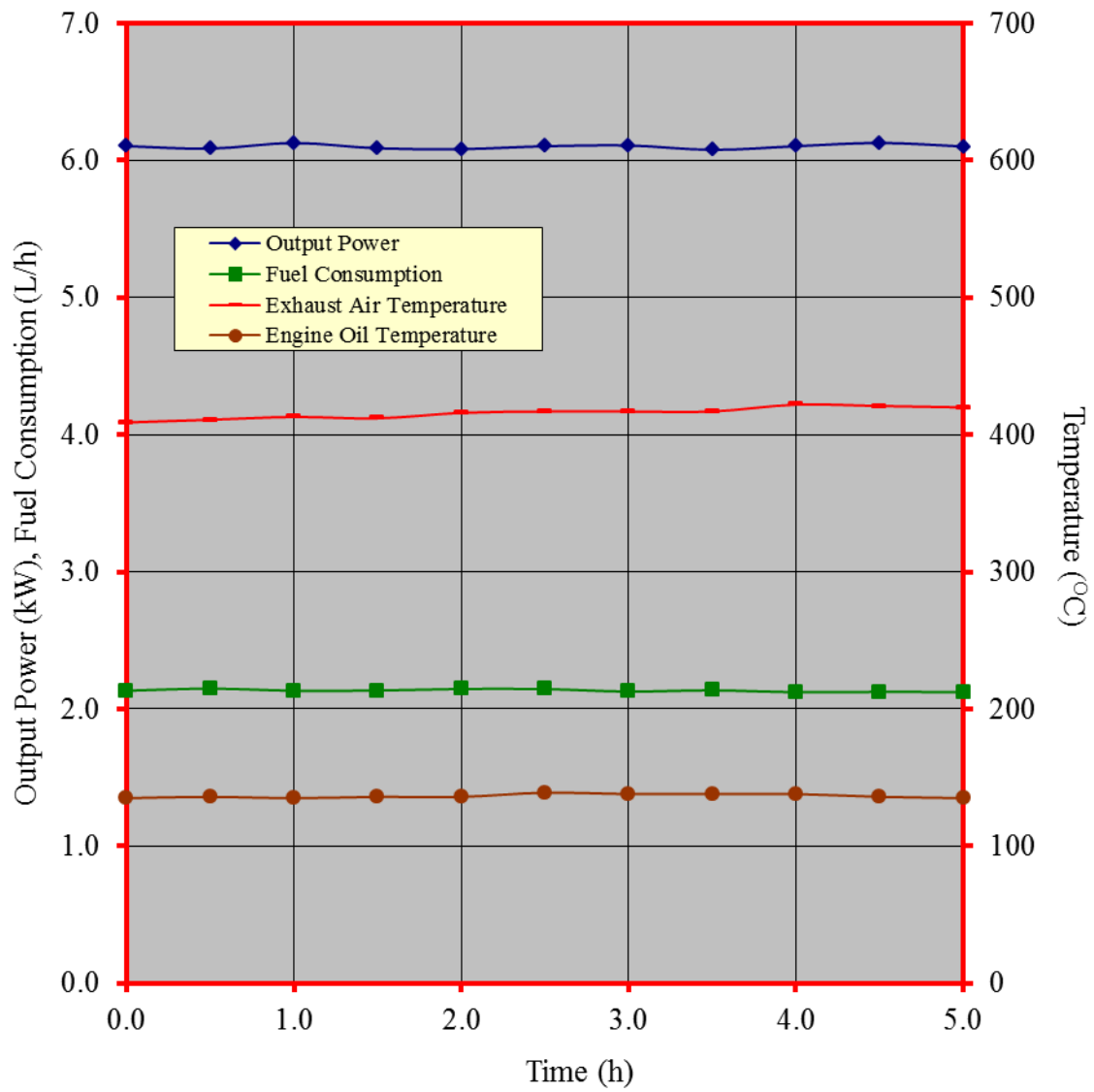


Figure 4. Performance characteristic curves during the continuous running test of the KOHLER KD440 diesel engine

DATA COMPUTATION

ANNEX. Formula used during calculation and testing

1. Varying Load Test

The engine was mounted on a dynamometer. The engine throttle was set at the maximum setting recommended by the manufacturer/test applicant. Data were gathered at different engine speeds starting from the maximum no-load speed down to the lowest speed. (The engine speed should not be lower than 600 rpm.) Data such as shaft torque; shaft speed; fuel consumption; temperatures of engine oil, exhaust air, and cooling water; vibration level at x, y, and z axis; and noise level were gathered at every speed setting. The following were computed:

- a. Output Power (kW), OP - The actual power at the output shaft of the engine is as given by equation 1.

$$OP = \frac{T \times N}{974} \quad , \quad \text{equation 1}$$

where:

OP = OutputPower, kW

T = Torque, kg - m

N = Shaft Speed, rpm

$$\text{Percent power} = \frac{\text{Actual maximum power(kW) on test}}{\text{Rated maximum power(kW) found on the plate/brochure}} \times 100$$

- b. Fuel Consumption (L/h) – It is the amount of fuel consumed by the engine per hour. This was obtained by volumetric method using ONO SOKKI Digital Fuel Consumption Meter (Model: FC-024) coupled to an ONO SOKKI Burette Stand (Model: PP-523).
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Annex, Con't.

- c. Specific Fuel Consumption (g/kW-h) - This is the amount of fuel consumed by the engine per unit of energy output. From test data, it was computed using equation 2.

$$SFC = \frac{FC \times D}{OP}, \quad \text{equation 2}$$

where:

SFC = Specific Fuel Consumption, g/kW - h

FC = Fuel Consumption, L/h

D = Density of Fuel used, g/L

- d. Temperature (°C) - These are the temperatures of the engine at different locations. They were measured using YOKOGAWA Digital Thermometer (Model: 2575) with Multi-Channel switch.

d.1 Engine Oil- the temperature of the engine oil at the crankcase during operation

d.2 Exhaust Air- the temperature of exhaust gas at the outlet of the muffler during operation

d.3 Cooling Water- the temperature of the water at the condenser during operation (if water cooled)

- e. Noise Level [db(A)]- The noise emitted by the engine 7.5 meters away from the muffler, 45° with respect to the axis of the engine shaft and the height of the microphone was 1.2 meters from the ground. **Table 5** shows the permissible exposure time for a certain level of sound emitted by the engine.

Table 5. Permissible exposure time for different sound levels

DURATION OF EXPOSURE PER DAY (h)	SOUND LEVEL [db(A)]
8	90
6	92
4	95
3	97
2	100
1.5	102
1	105
0.5	110
0.25 OR LESS	115

Based on "General Industry Guide for applying Safety and Health Standards",
29CFR 1910 OSHA, U.S. Dept. of Labor, 1973

Annex, Con't.

2. Continuous Running Test

This is the durability test for the engine. The engine was set to the recommended continuous power rating and at rated shaft speed found on the engine plate and/or brochure and was operated continuously for five hours. Data (same as in varying load test) were gathered every 30 minutes. (If the engine has no recommended continuous power, it will be set at 80% of the rated maximum power and at rated shaft speed.) During the test, the shaft speed of the engine was not set lower than the rated. The average of the five-hour test data were reported as continuous test data of the engine. The average actual power of the engine divided by the rated maximum power found on the plate/brochure was reported as the percent continuous power attained by the engine.

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